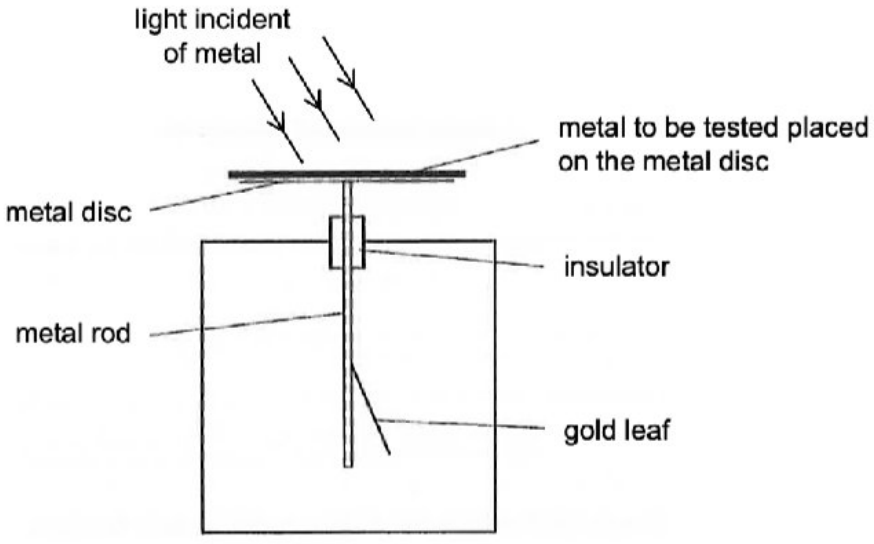
28 A student is given a gold-leaf electroscope and two light sources of wavelength 450 nm (blue light) and 650 nm (red light) to investigate the photoelectric effect of two metals – potassium and aluminium.

She intends to place the metal on the electroscope, neutralised the electroscope and shine one of the light on the metal. If the metal emits electrons, the gold leaf will rise as shown below.



The minimum energy to release electrons from the surface of the two metals are:

metal to be tested	minimum energy / eV
potassium	2.3
aluminium	4.3

Which of the following gives the correct observation when she conducts the investigation?

	potassium		aluminium	
	red light	blue light	red light	blue light
A	does not rise	rise	does not rise	rise
B	does not rise	rise	does not rise	does not rise
C	rise	does not rise	rise	rise
D	rise	does not rise	rise	does not rise